Jeffrey Wyatt-Wolfgang Marciniak-Haynosch

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EDUCATION

M.S. Microbiology | University of Illinois at Urbana-Champaign *Aug. 2021 – Sept. 2026*

Thesis: A Bacterial Guanylyl Cyclase Bridges Antiphage Response and Multiple Cellular Processes in Uropathogenic *Escherichia coli*.

B.S. General Biology | Eastern Connecticut State University *Aug. 2010 – Aug. 2014*

RESEARCH EXPERIENCE

Graduate Research Assistant | University of Illinois at Urbana-Champaign, Dept. of Microbiology *Jan. 2022 – Dec. 2024*

- Managed multiple concurrent research projects spanning computational, molecular biology, and structural biology workstreams; maintained meticulous records, tracked sample movement across the lab and core sequencing facilities, and consistently met experimental deadlines.
- Developed and optimized a high-sensitivity screening method to identify bacteriophages that trigger antiphage defense mechanisms in *E. coli*.
- Integrated next-generation sequencing with mutagenesis assays and ectopic-expression systems to identify phage-encoded proteins that antagonize bacterial antiphage defense.
- Utilized artificial intelligence to identify ambiguous protease cut sites.
- Performed *in silico* protein structure analyses using AlphaFold3 and ChimeraX; designed and validated functional fusion proteins through bioassays.
- Mentored undergraduate students, postdoctoral researchers, and research scientists in bacteriophage cultivation and classical phage characterization.

Research Associate | Armata Pharmaceuticals — Discovery & Engineering *May 2019 – Jun. 2020*

Supervisor: Dr. Sébastien Lemire

- Designed and implemented a unique-identifier system to track process-development trials and bacteriophage purification workflows across the discovery pipeline — directly analogous to scientific pipeline coordination work.
- Applied advanced molecular biology techniques to assemble plasmid constructs with therapeutic payloads for novel engineered phage development.
- Evaluated *Pseudomonas aeruginosa* phage-cocktail candidates in support of the SWARM-Pa Phase 1/2 clinical trial (ClinicalTrials.gov: NCT04596319).

Research Associate | C3J Therapeutics — Clinical Division *Jan. 2016 – May 2019*

Supervisors: Mr. Ken Lee, M.Sc. & Mr. Omid Sheikh, M.Sc.

- Assembled a comprehensive bacteriophage library screened against ESKAPE pathogens; tracked sample status and screening campaign documentation across multiple concurrent projects.
- Co-developed a universal diagnostic assay for detecting bacteriophage contamination in bioreactor systems.
- Conducted minimum inhibitory concentration (MIC) assays for a broad panel of synthetic peptides against commensal and pathogenic GI and oral bacteria.
- Contributed to the Phase 2 clinical study evaluating C16G2 Varnish and Strip (ClinicalTrials.gov: NCT03196219).

Laboratory Assistant | C3-Jian — Research & Development *Sept. 2015 – Jan. 2016*

Supervisor: Dr. Lucy Sim

- Maintained optimal laboratory conditions across departmental research activities, including equipment upkeep, reagent inventory, and waste management.

OPERATIONS & SYSTEMS EXPERIENCE

Operations & Systems Lead | Field of Dreams LLC, Higganum, CT *May 2025 – Present*

- Identified inefficiencies in field operations workflows and designed a structured property-assessment SOP for scoping, estimating, and crew handoffs across job sites.
- Streamlined truck-bed and trailer organization and standardized debris-removal procedures across job sites.

- Performed full-scope property assessments — mowing, garden-bed installation, edging, debris removal, and invasive-plant identification and remediation.
- Constructed dust-mitigation partition walls in a working horse barn, plus custom shelving and under-stairwell storage build-outs.

INDEPENDENT PROJECT

ScapeDaddy — Full-Stack Field-Operations Platform with an Intelligent Document-Processing Pipeline | *2026 – Present*
Self-directed

- Self-directed personal-time project, in active development. Designed and built **ScapeDaddy**, a Python/Flask + SQLite full-stack web application for independent landscaping and field-service businesses that turns messy end-of-day field notes into structured operations data; informed by hands-on field experience but developed independently of any employer.
- Engineered an LLM-driven intelligent document-processing pipeline that ingests unstructured, natural-language end-of-day field notes and converts them into validated, structured database updates (client records, work orders, labor hours, and payroll) behind a non-blocking human-review step.
- Adopted a plan-then-execute orchestration pattern over fragile native tool-calling to make multi-step database mutations from a hosted LLM backend (Groq) reliable, auditable, and reproducible.
- Implemented entity-resolution and progressive-enrichment logic that reconciles sparse, partial client records (e.g., first-name-only references) into confident matches without nagging for missing fields.
- Built a payroll engine supporting multiple pay models (revenue-split and hourly) with a day-window "true-hours" calculation, plus an internal cost-estimation ledger to formalize margin assumptions and standardize quote preparation.
- Implemented an idempotent database-migration helper to allow non-breaking schema evolution across deployments.
- Built bilingual (English/Spanish) and dual-unit (Imperial/Metric) interfaces to broaden workforce accessibility, applying the same documentation, tracking, and status-reporting principles used in scientific pipeline coordination.

TEACHING EXPERIENCE

Graduate Teaching Assistant — Introduction to Microbiology Lab | University of Illinois at Urbana-Champaign, Dept. of MCB *Spring 2023 – Spring 2024*

- Instructed 120-minute laboratory sessions for ~50 undergraduates twice weekly across two semesters, covering foundational microbiology techniques.
- Developed a standardized slide deck adopted department-wide by all Teaching Assistants.
- Proctored examinations, graded assignments, held weekly office hours, and authored letters of recommendation.

Graduate Teaching Assistant — Molecular & Cellular Basis of Life | University of Illinois at Urbana-Champaign, Dept. of MCB *Fall 2022*

- Facilitated four 50-minute discussion sections per week for ~25 undergraduates on core biology fundamentals.
- Proctored examinations, graded assignments, held weekly office hours, and authored letters of recommendation.

Undergraduate Teaching Assistant — Human Biology Laboratory | Eastern Connecticut State University, Dept. of Biology *Fall 2013 – Spring 2014*

- Delivered five 20-minute lectures on Human Anatomy and Physiology to ~20 students; prepared laboratory materials and equipment for each session.

POSTER PRESENTATIONS

Oct. 2024 **Marciniak-Haynosch JWW**, Farley KR, Vanderpool C, Huang R. "A guanylate cyclase has multiple roles in *Escherichia coli*: from bacteriophage defense to motility and cell wall synthesis." 44th Annual UIUC Microbiology Conference.

Oct. 2023 **Marciniak-Haynosch JWW**, Fatma S, Chakravarti A, Farley KR, Vanderpool C, Huang R. "Functional Analysis of Cyclic Oligonucleotide-Based Antiphage Signalling Systems." 43rd Annual UIUC Microbiology Conference.

Oct. 2022 **Marciniak-Haynosch JWW**, Fatma S, Chakravarti A, Farley KR, Vanderpool C, Huang R. "Functional Analysis of Cyclic Oligonucleotide-Based Antiphage Signalling Systems." 42nd Annual UIUC Microbiology Conference.

Aug. 2022 **Marciniak-Haynosch JWW**, Fatma S, Chakravarti A, Farley KR, Vanderpool C, Huang R. "Functional Analysis of Cyclic Oligonucleotide-Based Antiphage Signalling Systems." 2nd Annual Genomics and Eco-Evolution of Multi-Scale Interactions Symposium.

HONORS & AWARDS

2023 Graduate — Cold Spring Harbor Laboratory — Advanced Bacterial Genetics Course

2022 & 2023 Excellence in Teaching Award — University of Illinois at Urbana-Champaign

2013 Member — Beta Beta Beta (BBB) National Biology Honor Society

SERVICE & OUTREACH

Nov. 2022 **Judge** — 2022 Undergraduate Research Symposium, University of Illinois at Urbana-Champaign

Oct. 2022 **Organizer & Host** — 42nd Annual UIUC Microbiology Conference

Mar. 2022 **Member** — American Society for Microbiology

Aug. 2022 **Certified Personal Trainer** — National Association of Sports Medicine — Calisthenics, Functional Strengthening & Mobility

TECHNICAL SKILLS

Pipeline & Documentation Tools: Process-tracking systems (designed and implemented at Armata for phage purification trials); cross-team submission verification; structured documentation and SOP authorship; database design (SQLite) and migration tooling; relational schema design; spreadsheet, dashboard, and tracking-system fluency; willingness to learn specialized R&D tracking platforms.

Laboratory Techniques: Bacteria/Phage Isolation & Sequencing, CRISPRi Knockdowns, DNA Extraction, Gel Electrophoresis, Gibson Assembly, ICHIP Phage Isolation, λ RED Recombination, MIC Assays, P1 Transduction, PCR, Transposon Mutagenesis.

Bioinformatics: AlphaFold3, Autodock Vina, ChimeraX; long-read (Oxford Nanopore) and short-read phage genome assembly and alignment with Minimap2 (incl. via Geneious Prime), FastQC, Filtlong; NCBI and UniProt tools, NEBuilder, primer design, plasmid construction, variant calling.

Software & Computing: Python (Flask), R, SQL/SQLite, JavaScript, HTML/CSS; LLM application development (hosted inference APIs, multi-step LLM workflow orchestration [plan-then-execute], prompt engineering); Benchling, Geneious Prime, GraphPad Prism, Microsoft Office (Word/Excel/PowerPoint), Obsidian, Ubuntu (Unix/Linux).